LS-CAT Detector Comparison 1407

Detector	Eiger X 9M	Pilatus3 S 6M	Rayonix MX 300 HS	ADSC HF-4M
Delivery	12 Months	6 Months	12 Months	9 Months
Delivery Note		Production line has been delivering similar models.	1st ones were very late but it's likely that the major production bugs have been solved and time is conservative.	First one is about to be delivered. Quoted time is likely optimistic.
Mode	Photon Counting PAD	Photon Counting PAD	Integrating CCD	Counting or Integrating PAD mode selectable module by module
Active Area (mm x mm)	233.2 x 245.2	423.6 x 434.6	300 x 300	340 x 310
Gaps (percent of active area)	6.30%	8.50%	50 µm wide (0.1 % of active area)	8.5 %
# Edge Pixels	74,112 (0.7%)	81,840 (1.3%)	245,760 (1.7%) 2 x 2 binning	81,920 (2.0%)
Nominal Pixel Size (µm x µm)	75 x 75	172 x 172	1 x 1 bin: 39 x 39 2 x 2 bin: 78 x 78 3 x 3 bin: 117 x 117 4 x 4 bin: 156 x 156 etc to 10 x 10 binning	150 x 150
Nominal # Pixels	10,166,590	6,224,001	1 x 1 bin: 58,982,400 2 x 2 bin: 14,745,600 3 x 3 bin: 6,553,600 4 x 4 bin: 3,686,400 etc to 10 x 10 binning	4,194,304
Maximum Readout Rate (Hz)	238	25	bin High Speed Low Noise 1 x 1 3 1.5 2 x 2 10 6 3 x 3 20 12 4 x 4 35 20 etc. to 10 x 10 binning	200
Deadtime (ms)	0.003	2	1	0.001
Sensor	450 µm Si	450 µm Si	40 µm Gd2O2S	500 µm Si
QE	0.84	0.84	0.83	0.8
Read Noise	0	0	8 e/pixel (fast) 4 e/pixel (slow)	0 (counting) 0.2 x-ray (integrating)
Point Spread Function FWHM (µm	75	172	100	<150
Photons/Second per pixel @ 12 KeV	2E+06	1E+07	5E4 (fast mode) 1E5 (slow mode)	1E5 (counting w/ 1% loss) 1E8 (integrating)
			Note that diffraction spots in the typical 2x2 mode take up several pixels so these numbers should be increased by some factor, say 5, to compare with the Pilatus and ADSC which would typically measure the same data in one pixel.	